

DATA1901 Extreme Curveball Practice Exam C

Hard in-scope practice paper with answer key and worked solutions

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DATA1901 Extreme Curveball Practice Exam C

This is an intentionally difficult DATA1901-style practice paper. It is still within the scope of the local DATA1901 materials, but the questions are designed to be less direct than a standard paper. The goal is to solidify deep understanding by forcing you to connect ideas across study design, graphical summaries, numerical summaries, probability, models, inference, R code, and data-story interpretation.

This is not an official University of Sydney examination paper.

Curveball Warning

This paper is harder than the previous practice papers. It uses:

- boundary cases, such as equality at a boxplot threshold;
- wording traps, such as “prove”, “all”, “guarantee”, and “the null is true”;
- model-choice traps, such as binomial conditions and paired versus independent data;
- R interpretation traps, such as `lm(y ~ x)` direction and `pnorm` versus `qnorm`;

- data-story traps, where the output is real evidence but does not support the strongest claim.

The correct strategy is not memorisation. The correct strategy is:

1. Identify the variable types.
2. Identify the population, sample, parameter, and statistic.
3. Identify what the study design permits.
4. Identify the model assumptions.
5. Interpret in context without overclaiming.

Instructions

Time allowed: 10 minutes reading time, 2 hours writing time.

Section	Questions	Suggested weighting
Multiple Choice	30 questions	75%
Extended Answer	1 data-story question with 6 parts	25%

Choose at most one answer for each multiple choice question. For the extended answer, show reasoning and write conclusions in context.

Section A: Multiple Choice

In each question, choose the most appropriate option.

Integrated DATA1901 Concepts

1. A researcher randomly samples 2000 adults from NSW and asks whether they use a meal-delivery app. App users report higher weekly spending on takeaway food. Which conclusion is most defensible?
 - (a) App use causes higher takeaway spending because the sample was random
 - (b) The study can help estimate population features for NSW adults, but the association is not automatically causal
 - (c) The study is useless because there is no treatment
 - (d) The null hypothesis is true because no experiment was run
2. A study finds that Treatment A has a higher success rate than Treatment B overall, but within each age group Treatment B has a higher success rate. What idea is most directly involved?
 - (a) Simpson's paradox
 - (b) The Central Limit Theorem
 - (c) The 68-95-99.7 rule
 - (d) The binomial coefficient
3. A council releases a "de-identified" dataset containing location, age, rare medical condition, and date of hospital visit. Which concern is most appropriate?
 - (a) De-identification always removes all privacy risk
 - (b) Combining variables can still make re-identification possible
 - (c) The dataset cannot be tidy if names are removed
 - (d) Privacy matters only in randomised experiments
4. A dataset contains exam_mark (quantitative), faculty (qualitative), and campus (qualitative). Which display best compares exam marks across faculties while also showing campus?
 - (a) Histogram of campus, faceted by exam_mark
 - (b) Scatterplot of faculty against campus
 - (c) Comparative boxplots of exam_mark by faculty, with colour or faceting by campus
 - (d) Q-Q plot of faculty
5. Which code maps the variable species to colour in a scatterplot rather than setting every point to one fixed colour?
 - (a) `ggplot(dat, aes(x = length, y = width)) + geom_point(colour = species)`
 - (b) `ggplot(dat, aes(x = length, y = width, colour = species)) + geom_point()`
 - (c) `ggplot(dat, aes(x = species, colour = "blue")) + geom_point()`
 - (d) `ggplot(dat, aes(x = length, y = width)) + geom_point(aes(colour = "species"))`
6. A variable x has mean 15 and standard deviation 4. Define $y = 100 - 3x$. What are the mean and standard deviation of y ?
 - (a) Mean = 55, SD = -12
 - (b) Mean = 55, SD = 12
 - (c) Mean = 145, SD = 12
 - (d) Mean = 55, SD = 4
7. A dataset has $Q1 = 12$ and $Q3 = 20$. Using the standard boxplot rule, which statement is correct?
 - (a) The IQR is 32
 - (b) The lower threshold is 0 and the upper threshold is 32

- (c) A value exactly equal to 32 is always a high outlier
 (d) A value of -1 is not unusual
8. If $X \sim N(40, 5^2)$, which R command calculates $P(X < 35 \text{ or } X > 50)$?
- (a) `pnorm(35, 40, 5) + (1 - pnorm(50, 40, 5))`
 (b) `pnorm(50, 40, 5) - pnorm(35, 40, 5)`
 (c) `qnorm(35, 40, 5) + qnorm(50, 40, 5)`
 (d) `1 - pnorm(35, 40, 5) + pnorm(50, 40, 5)`
9. If $X \sim N(10, 2^2)$, which command finds the cutoff value with 5% of values above it?
- (a) `pnorm(0.95, 10, 2)`
 (b) `qnorm(0.05, 10, 2)`
 (c) `qnorm(0.95, 10, 2)`
 (d) `1 - pnorm(0.05, 10, 2)`
10. A machine has no systematic bias, but individual readings vary randomly. If the same object is measured 100 times and the average is reported, what mainly happens?
- (a) Chance error in the average tends to reduce, but systematic bias would not be fixed by averaging
 (b) Bias must increase because the sample size is larger
 (c) Standard deviation of individual readings becomes negative
 (d) The exact value becomes unknowable because there are too many readings
11. For a regression of y on x , suppose $\bar{x} = 10$, $\bar{y} = 50$, $SD_x = 4$, $SD_y = 8$, and $r = -0.5$. What is the fitted regression line?
- (a) $\hat{y} = 60 - 1 \cdot x$
 (b) $\hat{y} = 40 + 1 \cdot x$
 (c) $\hat{x} = 60 - 1 \cdot y$
 (d) $\hat{y} = 50 - 0.5 \cdot x$
12. A fitted line is $\hat{y} = 7 + 2x$. For an observation with $x = 4$ and observed $y = 12$, what is the residual?
- (a) 3
 (b) -3
 (c) 15
 (d) -15
13. Two variables have correlation $r = -0.30$. New variables are created by $u = 5 - 2x$ and $v = 10 - 4y$. What is the correlation between u and v ?
- (a) -0.30
 (b) 0.30
 (c) 0.60
 (d) -0.60
14. A rare disease affects 1 in 10,000 people. A screening test has a low false positive rate. A person tests positive. Why is it still dangerous to conclude immediately that the person almost certainly has the disease?
- (a) The mean and median must be equal first
 (b) The base rate of the disease matters for interpreting $P(\text{disease} \mid \text{positive})$
 (c) The Central Limit Theorem forbids screening tests
 (d) A low false positive rate proves all positives are false
15. Which situation is least suitable for a simple binomial model?

- (a) 20 independent coin tosses with the same chance of heads each toss
 - (b) 50 independent website visits where each visitor either clicks or does not click, with the same click probability
 - (c) Drawing 10 names without replacement from a class of 12 where exactly 3 students are scholarship students
 - (d) 30 independent free throws with the same chance of success on each throw
16. Let $X \sim \text{Bin}(12, 0.25)$. Which R command calculates $P(X \geq 4)$?
- (a) `pbinom(4, 12, 0.25)`
 - (b) `1 - pbinom(4, 12, 0.25)`
 - (c) `1 - pbinom(3, 12, 0.25)`
 - (d) `dbinom(4, 12, 0.25)`
17. A box has population mean 2 and population SD 5. You draw 100 times with replacement. Which statement is correct?
- (a) The sample mean has EV = 200 and SE = 50
 - (b) The sample sum has EV = 2 and SE = 0.5
 - (c) The sample sum has EV = 200 and SE = 50
 - (d) The sample mean has EV = 2 and SE = 50
18. Compared with sampling with replacement, sampling without replacement from a finite population:
- (a) usually increases standard error because there are fewer repeated values
 - (b) can reduce standard error when the sample is a non-negligible fraction of the population
 - (c) makes bias disappear automatically
 - (d) makes the Central Limit Theorem impossible
19. A two-sided test of $H_0: \mu = 100$ at $\alpha = 0.05$ corresponds to a 95% confidence interval for μ of (101.2, 108.5). What conclusion is most consistent?
- (a) Reject H_0 , because 100 is outside the interval
 - (b) Do not reject H_0 , because the interval is 95% confident
 - (c) Accept H_0 , because 100 is close to 101.2
 - (d) The p-value must be exactly 0.05
20. A one-sample t-test is used instead of a z-test mainly because:
- (a) the sample mean is qualitative
 - (b) the population standard deviation is unknown and estimated using the sample SD
 - (c) the null hypothesis is always false
 - (d) t-tests do not require independent observations

Data Story 1: Remote Work, Meetings, and Burnout

Use the following information to answer Questions 21-30.

A workplace analytics team studies 800 employees to investigate remote work, meeting load, burnout, and productivity.

```
dim(workplace)
```

```
[1] 800 15
```

```
names(workplace)
```

```
[1] "employee_id"      "department"
[3] "role_level"        "remote_days"
[5] "remote_group"      "meetings_per_week"
[7] "focus_hours"      "manager_training"
[9] "burnout_score"     "productivity_score"
[11] "promotion"         "tenure_years"
[13] "age_group"         "left_company"
[15] "team_size"
```

Variable	Description
employee_id	unique employee identifier
department	employee department
role_level	junior, mid, senior
remote_days	days per week worked remotely
remote_group	office, hybrid, or remote
meetings_per_week	number of meetings per week
focus_hours	uninterrupted work hours per week
manager_training	whether the employee's manager completed a training program
burnout_score	score from 1 to 10
productivity_score	score from 0 to 100
promotion	whether the employee was promoted this year
tenure_years	years employed
age_group	under 30, 30-49, 50 plus
left_company	whether the employee left the company
team_size	number of people in the team

Additional R output:

```
hybrid = workplace %>% filter(remote_group == "hybrid")
```

```
dim(hybrid)[1] / dim(workplace)[1]
```

```
[1] 0.36
```

```
lm(productivity_score ~ meetings_per_week, data = workplace)
```

Coefficients:

(Intercept) meetings_per_week

78.0 -1.1

```
cor(workplace$meetings_per_week, workplace$productivity_score)
[1] -0.34
```

```
t.test(burnout_score ~ manager_training, data = workplace)
```

Two Sample t-test

p-value = 0.041

sample estimates:

mean in group FALSE	mean in group TRUE
7.2	6.6

```
chisq.test(table(workplace$promotion, workplace$remote_group))
```

Pearson's Chi-squared test

p-value = 0.049

21. Which variable is most clearly an identifier?
 - (a) employee_id
 - (b) remote_days
 - (c) productivity_score
 - (d) department
22. Which pair is most suitable for a simple linear regression without first converting variables?
 - (a) department and promotion
 - (b) meetings_per_week and productivity_score
 - (c) remote_group and age_group
 - (d) employee_id and left_company
23. What does the output 0.36 tell us?
 - (a) 36 employees are hybrid
 - (b) 36% of employees are in the hybrid remote-work group
 - (c) Hybrid employees score 36 on productivity
 - (d) Hybrid employees have 0.36 meetings per week
24. Which graph would best compare burnout scores across the three remote-work groups and also show role level?
 - (a) Histogram of remote_group
 - (b) Scatterplot of remote_group against role_level
 - (c) Comparative boxplots of burnout_score by remote_group, coloured or faceted by role_level
 - (d) Q-Q plot of department
25. What is the predicted productivity score for an employee with 10 meetings per week using the fitted line?
 - (a) 67.0
 - (b) 76.9
 - (c) 89.0
 - (d) -11.0
26. Which interpretation of the regression slope is most appropriate?
 - (a) Each extra meeting per week is associated with a predicted 1.1 point lower productivity score
 - (b) Each extra productivity point causes 1.1 fewer meetings per week

- (c) The correlation is -1.1
 - (d) Every employee with no meetings has productivity exactly 78
27. The correlation between meetings and productivity is -0.34 . Which statement is best?
- (a) There is a perfect negative linear relationship
 - (b) There is a moderate negative linear association, not proof of causation
 - (c) Meetings are categorical
 - (d) The fitted slope must be exactly -0.34
28. The t-test p-value is 0.041. Which conclusion is most defensible at $\alpha = 0.05$?
- (a) There is evidence that mean burnout differs by manager-training group, with the trained-manager group lower in the sample
 - (b) Manager training definitely causes lower burnout
 - (c) 4.1% of employees have trained managers
 - (d) The null hypothesis is true because 0.041 is small
29. For the chi-squared test p-value 0.049, which conclusion changes if the analyst uses $\alpha = 0.01$ instead of $\alpha = 0.05$?
- (a) At 0.01, there is not enough evidence of association between promotion and remote group
 - (b) At 0.01, the evidence becomes stronger
 - (c) At 0.01, remote work is proven to cause promotion
 - (d) Alpha changes the observed data values
30. A report says: "Hybrid work causes promotion differences because the chi-squared p-value is 0.049." What is the main flaw?
- (a) Chi-squared tests cannot ever use categorical variables
 - (b) The p-value is a probability that the null is true
 - (c) A small p-value can indicate association, but the observational data do not prove causation
 - (d) The output should have used a histogram of `employee_id`

Section B: Extended Answer

There is one question in this section. This is intentionally harder than the standard extended answer. You should still answer using DATA1901 principles: evidence, context, variable types, assumptions, and careful language.

Data Story 2: Clinic Reminders, Waiting Time, and Patient Satisfaction

A health service is evaluating appointment systems across 60 clinics. The dataset `clinic_access` contains 1800 patient appointments.

Source 1: Data Dictionary

Variable	Description
<code>appointment_id</code>	unique appointment identifier
<code>clinic_id</code>	clinic identifier
<code>region</code>	metro, outer metro, or regional
<code>patient_age_group</code>	under 30, 30-64, 65 plus
<code>appointment_type</code>	standard, urgent, or follow-up
<code>sms_reminder</code>	whether an SMS reminder was sent
<code>booked_days_ahead</code>	days between booking and appointment
<code>wait_min</code>	waiting time at clinic in minutes
<code>no_show</code>	whether the patient did not attend
<code>satisfaction_score</code>	patient satisfaction from 1 to 10
<code>clinic_training</code>	whether the clinic completed access-improvement training
<code>period</code>	before or after the reminder program

Source 2: Paired before/after output

For each of the 60 clinics, the average monthly no-show percentage was calculated before and after the SMS reminder program.

```
t.test(after_no_show_pct, before_no_show_pct, paired = TRUE)
```

Paired t-test

```
t = -2.38, df = 59, p-value = 0.0205
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -3.9 -0.3
sample estimates:
mean difference
 -2.1
```

Source 3: Chi-squared test

```
chisq.test(table(clinic_access$sms_reminder, clinic_access$no_show))
```

Pearson's Chi-squared test

```
X-squared = 7.84
```

```
df = 1
p-value = 0.0051
```

Source 4: Linear model

```
summary(lm(satisfaction_score ~ wait_min, data = clinic_access))
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.42	0.11	76.55	<0.001
wait_min	-0.052	0.006	-8.67	<0.001

Residual standard error: 1.6

Multiple R-squared: 0.17

Source 5: Additional warning from the analyst

Clinics adopted the reminder program at different times. During the same period, some clinics also changed staffing levels, appointment triage rules, and opening hours. The supplied outputs do not adjust for these other changes.

Question 1

(a) A draft report says:

“SMS reminders caused no-shows to fall in every clinic, and shorter waiting times caused all patients to become satisfied. The effect is especially strong for patients aged 65 plus.”

Explain three problems with this statement using Sources 2-5.

(b) Write a correct interpretation of Source 2. Your answer must mention why the test is paired.

(c) A student explains Source 3 by saying:

“The p-value is 0.0051, so there is a 0.51% chance that SMS reminders and no-shows are independent.”

Write a corrected explanation.

(d) The health service wants to investigate whether the relationship between waiting time and satisfaction differs by appointment type. Describe an appropriate plot and write R code that could produce it.

(e) Suppose the analyst wants to test whether mean waiting time differs between clinics that completed access-improvement training and clinics that did not. Identify the response variable, explanatory variable, and an appropriate test.

(f) Suggest one additional research question using three variables from Source 1. Briefly explain what graphical summary or statistical method would be appropriate.

Practice Exam Guide

Numerical Summaries and Transformations

Idea	Formula or rule
IQR	$IQR = Q3 - Q1$
Boxplot lower threshold	$LT = Q1 - 1.5 \cdot IQR$
Boxplot upper threshold	$UT = Q3 + 1.5 \cdot IQR$
Linear transformation	If $y = a + b \cdot x$, then $\text{mean}(y) = a + b \cdot \text{mean}(x)$ and $SD(y) = \text{abs}(b) \cdot SD(x)$

Models

Model	Useful fact
Normal	$X \sim N(\text{mean}, SD^2)$
Regression slope	$b = r \cdot SD_y / SD_x$
Regression intercept	$a = \bar{y} - b \cdot \bar{x}$
Residual	observed y - predicted y
Binomial	Fixed n , binary outcome, constant p , independent trials
Box model sample sum	$EV = n \cdot M, SE = \sqrt{n} \cdot SD$
Box model sample mean	$EV = M, SE = SD / \sqrt{n}$

Probability

Rule	Formula
Addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Multiplication rule	$P(A \text{ and } B) = P(A) \cdot P(B A)$
Binomial exact probability	$\text{choose}(n, x) \cdot p^x \cdot (1-p)^{(n-x)}$
Binomial cumulative probability	$\text{pbinom}(x, n, p)$

R Commands

```
library(tidyverse)
ggplot(data, aes(x = quantitative_var)) + geom_histogram()
ggplot(data, aes(x = group_var, y = quantitative_var)) + geom_boxplot()
ggplot(data, aes(x = x_var, y = y_var)) + geom_point()
ggplot(data, aes(x = x_var, y = y_var, colour = group_var)) + geom_point()
pnorm(value, mean, sd)
qnorm(probability, mean, sd)
pbinom(x, size, prob)
lm(y ~ x, data = data)
t.test(y, mu = constant)
t.test(y ~ group, data = data)
t.test(after, before, paired = TRUE)
chisq.test(table(var1, var2))
```

Answer Key

Multiple Choice Answers

Question	Answer	Question	Answer	Question	Answer
1	B	11	A	21	A
2	A	12	B	22	B
3	B	13	A	23	B
4	C	14	B	24	C
5	B	15	C	25	A
6	B	16	C	26	A
7	B	17	C	27	B
8	A	18	B	28	A
9	C	19	A	29	A
10	A	20	B	30	C

Worked Solutions

Multiple Choice Worked Solutions

Q1. Answer: B

A random sample helps with generalising to the population it represents. It is not random assignment to app use, so the observed association does not prove app use causes higher spending.

Q2. Answer: A

This is Simpson's paradox: an overall association reverses or changes after accounting for a grouping variable.

Q3. Answer: B

Removing names does not guarantee privacy. A rare combination of location, age, condition, and date may still identify someone.

Q4. Answer: C

exam_mark is quantitative, while faculty and campus are qualitative. Comparative boxplots can compare marks by faculty, and colour or faceting can show campus.

Q5. Answer: B

Putting `colour = species` inside `aes()` maps the variable `species` to colour. Setting colour outside `aes()` makes it a fixed aesthetic.

Q6. Answer: B

The mean is $100 - 3 \times 15 = 55$. The SD is $\text{abs}(-3) \times 4 = 12$. Standard deviation cannot be negative.

Q7. Answer: B

$\text{IQR} = 20 - 12 = 8$. The lower threshold is $12 - 1.5 \times 8 = 0$; the upper threshold is $20 + 1.5 \times 8 = 32$. A value exactly equal to a threshold is on the fence; strict outliers are beyond the fence.

Q8. Answer: A

The probability outside the interval is the lower tail below 35 plus the upper tail above 50: $\text{pnorm}(35, 40, 5) + (1 - \text{pnorm}(50, 40, 5))$.

Q9. Answer: C

If 5% of values are above the cutoff, then 95% are below it. Use $\text{qnorm}(0.95, 10, 2)$.

Q10. Answer: A

Averaging repeated unbiased measurements reduces chance error in the average. Averaging would not remove systematic bias if bias were present.

Q11. Answer: A

The slope is $b = r \times \text{SD}_y / \text{SD}_x = -0.5 \times 8 / 4 = -1$. The intercept is $a = \bar{y} - b \times \bar{x} = 50 - (-1 \times 10) = 60$. So $\hat{y} = 60 - x$.

Q12. Answer: B

The predicted value is $7 + 2 \times 4 = 15$. The residual is $\text{observed} - \text{predicted} = 12 - 15 = -3$.

Q13. Answer: A

Both variables are multiplied by negative constants. Reversing both axes preserves the sign of correlation, so the correlation remains -0.30 .

Q14. Answer: B

For rare conditions, the base rate matters. A low false positive rate alone does not determine the probability of disease given a positive result.

Q15. Answer: C

Simple binomial models require independent trials with constant success probability. Drawing without replacement from a small class changes probabilities from draw to draw.

Q16. Answer: C

$P(X \geq 4) = 1 - P(X \leq 3)$, so the correct command is `1 - pbinom(3, 12, 0.25)`.

Q17. Answer: C

For a sample sum, $EV = n \cdot M = 100 \cdot 2 = 200$, and $SE = \sqrt{n} \cdot SD = 10 \cdot 5 = 50$.

Q18. Answer: B

Sampling without replacement can reduce standard error when the sample is a substantial fraction of the finite population, because draws are less variable once units cannot reappear.

Q19. Answer: A

For a two-sided test at alpha 0.05, if the 95% confidence interval excludes the null value 100, reject H_0 .

Q20. Answer: B

A t-test is used for mean inference when the population SD is unknown and the sample SD is used to estimate it.

Q21. Answer: A

`employee_id` uniquely labels employees. It is an identifier, not a substantive measurement.

Q22. Answer: B

Simple linear regression uses two quantitative variables. `meetings_per_week` and `productivity_score` are suitable quantitative variables.

Q23. Answer: B

The code filters to hybrid employees and divides that count by the total number of employees. 0.36 means 36% are hybrid.

Q24. Answer: C

Burnout score is quantitative, while remote group and role level are qualitative. Comparative boxplots by remote group, coloured or faceted by role level, match the variable types.

Q25. Answer: A

The prediction is $78.0 - 1.1 \cdot 10 = 67.0$.

Q26. Answer: A

The slope says each additional meeting per week is associated with a predicted 1.1 point decrease in productivity score. Causal wording is not justified.

Q27. Answer: B

A correlation of -0.34 indicates a moderate negative linear association. It is not a perfect relationship and does not prove causation.

Q28. Answer: A

The p-value 0.041 is less than 0.05, so there is evidence of different mean burnout scores. The sample means show lower burnout in the trained-manager group.

Q29. Answer: A

At alpha 0.05, 0.049 would lead to rejecting independence. At alpha 0.01, 0.049 is not small enough, so there is not enough evidence at the stricter level.

Q30. Answer: C

The chi-squared test can provide evidence of association between categorical variables. It does not prove that hybrid work causes promotion differences.

Extended Answer Model Solution

1(a) Three problems with the draft report

The draft report says:

“SMS reminders caused no-shows to fall in every clinic, and shorter waiting times caused all patients to become satisfied. The effect is especially strong for patients aged 65 plus.”

Problem 1: It overclaims causation.

Source 2 and Source 3 provide evidence that no-show outcomes changed or are associated with SMS reminders. However, Source 5 warns that clinics adopted the reminder program at different times and also changed staffing levels, triage rules, and opening hours. These are possible confounders. The outputs do not prove that SMS reminders alone caused the change.

Problem 2: It says “every clinic”.

Source 2 reports an average paired difference across 60 clinics. The mean difference is -2.1 percentage points. That does not mean every individual clinic had a decrease.

Problem 3: It says shorter waiting times caused all patients to become satisfied.

Source 4 shows a negative association between `wait_min` and `satisfaction_score`: longer waits are associated with lower satisfaction scores on average. But this does not prove causation, and $R^2 = 0.17$ means waiting time explains only part of the variation in satisfaction.

Problem 4: It says the effect is especially strong for patients aged 65 plus.

None of Sources 2-5 analyse `patient_age_group` or an interaction involving age group. The age-specific claim is unsupported.

An exemplary concise answer:

The report overclaims. The before/after paired test and chi-squared test show evidence of changes or associations involving no-shows, but Source 5 lists other simultaneous clinic changes, so SMS reminders alone are not proven causal. Source 2 gives an average change across clinics, not a decrease in every clinic. Source 4 shows an average association between waiting time and satisfaction, not that shorter waits cause all patients to become satisfied. The claim about patients aged 65 plus is unsupported because age group is not analysed in the outputs.

1(b) Correct interpretation of Source 2

Source 2 is a paired t-test because the before and after no-show percentages are matched within the same clinics. Each clinic contributes a before value and an after value, so the natural unit of analysis is the within-clinic difference.

The null hypothesis is that the mean before/after difference is zero. The output reports the mean difference as `after - before = -2.1`, with p-value 0.0205 and a 95% confidence interval from -3.9 to -0.3.

At the 5% level, there is evidence that the mean clinic no-show percentage changed after the reminder program. The negative mean difference suggests the average no-show percentage was lower after the program. However, because of Source 5, this should not be stated as definitive causal proof that SMS reminders alone caused the reduction.

1(c) Correct explanation of Source 3

Source 3 is a chi-squared test of independence between `sms_reminder` and `no_show`.

The p-value is 0.0051. This means that if SMS reminder status and no-show status were truly independent in the population, data at least this inconsistent with independence would be unlikely.

Since $0.0051 < 0.05$, there is evidence of an association between SMS reminder status and no-show status in this dataset.

The p-value is not the probability that the null hypothesis is true. It is not a 0.51% chance that SMS reminders and no-shows are independent.

1(d) Appropriate plot and R code

The question asks whether the relationship between waiting time and satisfaction differs by appointment type.

Variables:

- wait_min: quantitative explanatory variable
- satisfaction_score: quantitative response variable
- appointment_type: qualitative grouping variable

An appropriate plot is a scatterplot of satisfaction against waiting time, coloured by appointment type.

```
library(tidyverse)

ggplot(clinic_access,
      aes(x = wait_min,
          y = satisfaction_score,
          colour = appointment_type)) +
  geom_point() +
  labs(
    title = "Satisfaction by waiting time and appointment type",
    x = "Waiting time at clinic (minutes)",
    y = "Patient satisfaction score"
  )
```

A stronger exploratory version could add separate linear smooths by appointment type:

```
ggplot(clinic_access,
      aes(x = wait_min,
          y = satisfaction_score,
          colour = appointment_type)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Satisfaction by waiting time and appointment type",
    x = "Waiting time at clinic (minutes)",
    y = "Patient satisfaction score"
  )
```

1(e) Response, explanatory variable, and appropriate test

Question:

Does mean waiting time differ between clinics that completed access-improvement training and clinics that did not?

Response variable: wait_min, which is quantitative.

Explanatory variable: clinic_training, which is qualitative with two groups.

Appropriate test: a two-sample t-test comparing mean waiting time between the training and non-training groups, assuming independent observations or an appropriate unit of analysis. A careful

answer would also mention that if many patients come from the same clinic, independence at the patient level may be questionable, because observations within a clinic could be related.

Example R code:

```
t.test(wait_min ~ clinic_training, data = clinic_access)
```

1(f) Additional research question and method

Example research question:

Is the relationship between booked days ahead and no-show status different across appointment types?

Variables:

- booked_days_ahead: quantitative
- no_show: qualitative/binary
- appointment_type: qualitative

Suitable graphical summary:

Use comparative boxplots of booked_days_ahead by no_show, faceted or coloured by appointment_type.

```
ggplot(clinic_access,
       aes(x = no_show,
           y = booked_days_ahead,
           fill = appointment_type)) +
geom_boxplot() +
labs(
  title = "Booking lead time by no-show status and appointment type",
  x = "No-show status",
  y = "Days booked ahead"
)
```

Alternative valid research question:

Are no-show status and region associated, and does the pattern differ by appointment type?

This could begin with contingency tables or segmented bar charts of no_show by region, faceted by appointment_type.

Marking Guide for Extended Answer

Part	Marks	Examiner expectations
1(a)	9	Identifies three genuine problems; uses Sources 2-5; distinguishes average from every case; identifies unsupported age-group claim; avoids causal overclaim.
1(b)	5	Explains paired structure; interprets mean difference, CI, p-value; concludes in context; notes causation limitation.
1(c)	4	Correctly interprets p-value under the null; states association conclusion; rejects “probability null is true” wording.
1(d)	5	Correct variable-type reasoning; suitable scatterplot; correct ggplot code; uses colour/facet for appointment type.
1(e)	4	Identifies response/explanatory variables; chooses two-sample t-test; notes independence or clustering issue for a stronger answer.
1(f)	3	Gives coherent three-variable question and matching graph/method.

Why This Paper Is Hard

Difficulty feature	Example
Random sampling is not random assignment	Q1
Overall pattern can hide subgroup pattern	Q2
De-identification is not absolute privacy	Q3
Aesthetic mapping in ggplot	Q5
Boundary of boxplot thresholds	Q7
Upper/lower tails in R	Q8-Q9, Q16
Regression from summary statistics	Q11
Residual sign	Q12
Correlation under negative transformations	Q13
Base-rate reasoning	Q14
Binomial assumption failure	Q15
CI/test duality	Q19
Alpha sensitivity	Q29
Association versus causation in real outputs	Q28-Q30, Extended Answer

How To Use This Paper

After marking it, do not just count your score. For every missed question, classify the mistake:

Mistake type	What to revise
Variable type error	Topic 2 and test-selection maps
Formula substitution error	Topic 3, Topic 4, Topic 7
Tail-direction error	Topic 4, Topic 6, Topic 9
Overclaiming causation	Topic 1, Topic 5, Topic 8
R syntax interpretation	Coding milestones and ggplot reference
p-value or CI wording error	Topic 8, Topic 9, Topic 10, Topic 11
Confusing SD and SE	Topic 7